

# DBMS: ASSIGNMENT 6

## Database Triggers

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### Case Study: Staff Salary Audit & Tax Trigger Suite

```
mysql> USE StaffSalaryAuditSystem;  
Database changed
```

```
mysql> CREATE TABLE Staff (  
-> Id INT PRIMARY KEY,  
-> Name VARCHAR(50),  
-> Salary INT,  
-> Performance_Score INT  
-> );  
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> CREATE TABLE salary_audit_log (  
-> log_id INT AUTO_INCREMENT PRIMARY KEY,  
-> Id INT,  
-> Action VARCHAR(20),  
-> old_salary INT,  
-> new_salary INT,  
-> log_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
-> );  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Staff VALUES  
-> (1,'Adi Maitre',10000,5),  
-> (2,'Kritika Chaudhari',15000,4),  
-> (3,'Ritobrata Baksi',20000,3),  
-> (4,'Soham Kulkari',18000,2);  
Query OK, 4 rows affected (0.01 sec)
```

```
mysql> SELECT * FROM Staff;
```

| Id | Name              | Salary | Performance_Score |
|----|-------------------|--------|-------------------|
| 1  | Adi Maitre        | 10000  | 5                 |
| 2  | Kritika Chaudhari | 15000  | 4                 |
| 3  | Ritobrata Baksi   | 20000  | 3                 |
| 4  | Soham Kulkari     | 18000  | 2                 |

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```
mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_after_insert_staff
-> AFTER INSERT ON Staff
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO salary_audit_log(Id, Action, old_salary, new_salary)
-> VALUES(NEW.Id,'INSERT',NULL,NEW.Salary);
-> END$$
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
```

```
mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_after_update_staff
-> AFTER UPDATE ON Staff
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO salary_audit_log(Id, Action, old_salary, new_salary)
-> VALUES(OLD.Id,'UPDATE',OLD.Salary,NEW.Salary);
-> END$$
Query OK, 0 rows affected (0.02 sec)
mysql> DELIMITER ;
```

```
mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_after_delete_staff
-> AFTER DELETE ON Staff
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO salary_audit_log(Id, Action, old_salary, new_salary)
-> VALUES(OLD.Id,'DELETE',OLD.Salary,NULL);
-> END$$
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
```

```
mysql> INSERT INTO Staff VALUES (5,'Sarang Kolekar',25000,4);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> UPDATE Staff
-> SET Salary = 30000
-> WHERE Id = 1;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> DELETE FROM Staff WHERE Id = 2;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> SELECT * FROM salary_audit_log;
```

| log_id | Id | Action | old_salary | new_salary | log_date            |
|--------|----|--------|------------|------------|---------------------|
| 1      | 5  | INSERT | NULL       | 25000      | 2026-04-14 11:53:58 |
| 2      | 1  | UPDATE | 10000      | 30000      | 2026-04-14 11:54:16 |
| 3      | 2  | DELETE | 15000      | NULL       | 2026-04-14 11:54:31 |

Project: Smart Income Tracking & Audit Trigger System

```
mysql> USE SmartIncomeAuditSystem;
```

Database changed

```
mysql> CREATE TABLE user_login (
```

```
-> user_id BIGINT AUTO_INCREMENT PRIMARY KEY,
```

```
-> first_name VARCHAR(100),
```

```
-> last_name VARCHAR(100),
```

```
-> username VARCHAR(100),
```

```
-> role VARCHAR(20)
```

```
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> CREATE TABLE taxpayer_profile (
```

```
-> taxpayer_id BIGINT AUTO_INCREMENT PRIMARY KEY,
```

```
-> user_id BIGINT,
```

```
-> pan_number VARCHAR(10),
```

```
-> city VARCHAR(50)
```

```
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> CREATE TABLE income_sources (
```

```
-> income_id BIGINT AUTO_INCREMENT PRIMARY KEY,
```

```
-> taxpayer_id BIGINT,
```

```
-> source_type VARCHAR(50),
```

```
-> amount DECIMAL(10,2)
```

```
-> );
```

Query OK, 0 rows affected (0.04 sec)

```
mysql> CREATE TABLE income_audit_logs (
```

```
-> log_id INT AUTO_INCREMENT PRIMARY KEY,
```

```
-> user_id BIGINT,
```

```
-> action VARCHAR(50),
```

```
-> old_value VARCHAR(100),
-> new_value VARCHAR(100),
-> log_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
-> );
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO user_login VALUES
-> (1,'Adi','Maitre','adi123','taxpayer'),
-> (2,'Kritika','Chaudhari','kritika123','taxpayer');
Query OK, 2 rows affected (0.01 sec)
```

```
mysql> INSERT INTO taxpayer_profile VALUES
-> (1,1,'PAN001','Pune'),
-> (2,2,'PAN002','Mumbai');
Query OK, 2 rows affected (0.01 sec)
```

```
mysql> INSERT INTO income_sources VALUES
-> (1,1,'Salary',600000),
-> (2,2,'Business',300000);
Query OK, 2 rows affected (0.01 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_income_insert
-> AFTER INSERT ON income_sources
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO income_audit_logs(user_id, action, old_value, new_value)
-> VALUES(NEW.taxpayer_id,'INSERT',NULL,NEW.amount);
-> END$$
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
```

```
mysql> DELIMITER $$
mysql> CREATE TRIGGER trg_income_update
-> AFTER UPDATE ON income_sources
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO income_audit_logs(user_id, action, old_value, new_value)
-> VALUES(OLD.taxpayer_id,'UPDATE',OLD.amount,NEW.amount);
-> END$$
Query OK, 0 rows affected (0.02 sec)
mysql> DELIMITER ;
```

```
mysql> DELIMITER $$
```

```
mysql> CREATE TRIGGER trg_income_delete
-> AFTER DELETE ON income_sources
-> FOR EACH ROW
-> BEGIN
-> INSERT INTO income_audit_logs(user_id, action, old_value, new_value)
-> VALUES(OLD.taxpayer_id,'DELETE',OLD.amount,NULL);
-> END$$
Query OK, 0 rows affected (0.01 sec)
mysql> DELIMITER ;
```

```
mysql> INSERT INTO income_sources VALUES (3,1,'Freelance',200000);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> UPDATE income_sources
-> SET amount = 700000
-> WHERE income_id = 1;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> DELETE FROM income_sources
-> WHERE income_id = 2;
Query OK, 1 row affected (0.02 sec)
```

```
mysql> SELECT * FROM income_audit_logs;
+-----+-----+-----+-----+-----+-----+
| log_id | user_id | action | old_value | new_value | log_time          |
+-----+-----+-----+-----+-----+-----+
| 1      | 1       | INSERT | NULL      | 200000.00 | 2026-04-14 12:00:09 |
| 2      | 1       | UPDATE | 600000.00 | 700000.00 | 2026-04-14 12:00:18 |
| 3      | 2       | DELETE | 300000.00 | NULL      | 2026-04-14 12:00:26 |
+-----+-----+-----+-----+-----+-----+
```